

Current Hybrid Architecture Performance - Fifth Set

Version 10 | Hybrid Semantic Architecture | Fifth Set Benchmark

MODEL_VERSION: 10.11.22

GATE: P2610E1_FIFTH_SET_HYBRID_ARCHITECTURE_ACCURACY_BENCHMARK_V1_0

DECISION: PASS_WITH_LIMITATIONS

MODE: OFFLINE_REPLAY

Hybrid semantic authority: Claude Vision preferred after validation.

Deterministic engineering authority: geometry, cut length, development length, spacers, stirrup engineering, steel calculation.

This report is not a production-readiness claim.

EXECUTIVE SUMMARY

Beam identification: 76.47%

Bar identification: 36.68%

Correct of detected bars: 35.58%

Diameter identification: 74.34%

Steel accuracy: 48.00%

Overall accuracy: 49.18%

GT beams: 187 Detected: 143

GT bars: 1456 Identified: 534

MATCH bars: 190

Hybrid kg: 28219.705

Benchmark kg: 58796.332

Absolute kg error: 30576.627

FORMULAS (QA.2A / QA.3.0)

Beam ID = $\text{detected_gt_beams} / \text{total_gt_beams} * 100$

Bar ID = $\text{identified_gt_bars} / \text{total_gt_bars} * 100$

Correct of detected = $\text{MATCH} / \text{detected_bars} * 100$

Steel = $\max(0, 100 - \text{abs}(\text{hybrid-bench})/\text{bench} * 100)$

Overall = $\text{mean}(\text{beam ID, bar ID, correct of detected, steel})$

Diameter is reported separately and is not in the overall mean.

Vision usable beams (offline replay): 0

Vision scanned: 73 api_failed: 16

SCOPE

Fifth Set only. No Second/Third/Fourth/Sixth comparison.

No historical improvement claim. No production promotion.